POKHARA UNIVERSITY

Level : Bachelor | Semester – Spring | Year : 2020

Program: BE | Full Marks : 70

Course : Artificial Intelligence | Pass Marks: 31.5

Time : 2 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

Section– A: (5×10 = 50)

1. "Rational Thinking" as an aspect of Artificial Intelligence (AI), explain. Discuss on the various application domains of AI. 5+5

2. Mention search criteria. How are informed search methods more efficient than uninformed search? Explain the algorithm of A* search with a suitable example. 2+2+6

3. Discuss Propositional logic with its complexity, validity and tautology. Convert the following sentences into FOPL sentences. 5+5
   a) No mango is blue.
   b) You can fool some of the people all the times
   c) Everyone is younger than his father.
   d) None of the gardener likes sun.
   e) All purple mushrooms are venomous.

4. Differentiate Semantic net and Frame. Construct a parse tree for the sentence *"The students took the online examination"*. 4+6

5. Differentiate Supervised Learning with Unsupervised learning. Discuss K-means clustering with following dataset with k=2. 4+6

StaffID	Age	NO of leaves
101	20	4
102	29	7
103	35	6
104	22	9
105	28	8

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OR

Define Neural Network (NN). How back propagation algorithm is used in learning and predicting data in machine learning? Explain with suitable example. 2+8

Section – B: (1×20 = 20)

6. a) What do you mean by production system? Solve the following Crypto arithmetic problem. 2+8

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  COCA
+ COLA
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 PEPSI
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b) Introduce statistical reasoning and state Bayes Theorem. 4+4

A Scam detector system detects spam emails. Assume that the word "offer" occurs in 80% of the spam messages in your account and in 10% of your non-spam emails (i.e. genuine). 30% of the received emails are considered as a scam. Now, you will receive a new message which contains the word "offer". What is the probability that it is a spam?

1.

2.

3.

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